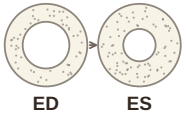
a Estimation: cine provides cues, correspondence is inferred, strain is defined by convention

1 · OBSERVED

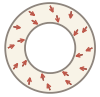
Cine bSSFP, short axis



boundaries visible; weak intramural texture; no material markers

I **2–3 · INFERRED**

Estimated displacement u(X, t)



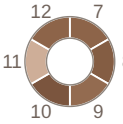
prior αR: tracking, registration, biomechanics, learned

registration example: weak cues can leave the map ambiguous; priors, supervision differ by method (Fig. 2)

$$\varphi^{\wedge} = \arg\min D(I_0, I_t \circ \varphi) + \alpha R(\varphi)$$

E R **4 · DEFINED**

Green–Lagrange Ecc, pixelwise



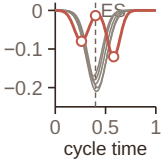
fill = Ecc; seg. 11 reduced

01 ED reference
02 E = ½(FᵀF − I)
03 c/r/l axes, signed
04 endo·mid·epi, 2D
05 ∂u/∂X, smoothed

choices 01–05 fix the value before any region is reported

5 · REPORTED

Segmental Ecc(t), mid ventricle



06 AHA mid 7–12
07 at ES **−0.01**
peak sys. **−0.08**
post-sys. **−0.12**
08 min(mean) **−0.16**
mean(min) **−0.18**

06–08 select one value from one curve (○, seg. 11)

Error sources can coexist; controlled diagnostic tests: **I** **input sufficiency** · re-image the motion finely **E** **estimation** · vary α; retrain if required **R** **representation** · fit known field; check strain too

b Validation: complementary endpoints; values as reported

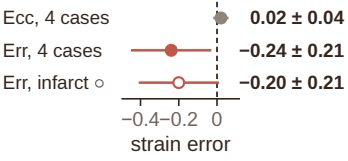
DENSE · tagging · cine FT · cine DL · ○ comparator · ✓ supported · ⊗ not established alone

Known motion

MRXCAT2.0 test of DeepStrain

✓ technical error
⊗ in-vivo accuracy

estimator error (mean ± SD)



○ infarct case, EF 49%. Dice 0.82; displacement 1.0 ± 0.9 mm. Slice means only (Fig. 4).

Paired DENSE / tagging

Militaru 2021; Cao 2018; Wang 2023

✓ material-sensitive agreement
⊗ reference error

ICC vs tagging

LS / CS / RS: FT–tagging



StrainNet (●) vs FT (○), Ecc



Militaru: n = 61, 3 T, RS range vs tagging by product. Cao: DENSE vs tagging Ecc 0.778. StrainNet (DL): 59 cine tests, DENSE reference.

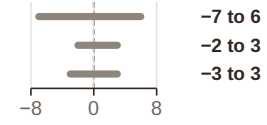
Agreement / repeatability

Yoon 2023; Bell 2025

✓ distinct measurement properties
⊗ accuracy (bias)

REGAIN vs GRAPPA

95% LoA, global FT strain



Strain-8 scan–rescan, 20 volunteers

global	**fair–excellent**
segmental	**more variable**

REGAIN: 49 participants; mean global differences near zero. Strain-8: same day, 8 protocols.

Tissue (LGE)

Kihlberg 2020

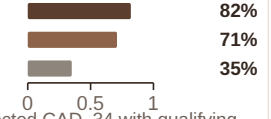
✓ scar-associated dysfunction
⊗ motion error

segments with LGE > 50%

AUC: end-systolic Ecc



sensitivity at 80% specificity



116 suspected CAD, 34 with qualifying scar. Ranks scar sensitivity, not motion accuracy.

Outcome & readers

Bello 2019; Masutani 2023

✓ prognostic / diagnostic link
⊗ clinical utility

C-index and AUC differ

Bello · RV motion in CTEPH, corrected C-index; n = 302



○ manual-variable model

Masutani · DL strain, WMA vs 4-reader consensus, AUC



DLSS: 53 patients, 846 segments. AUC spans thresholds; 30% Err is one operating point.